Aniketh Parkala

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Research Interests

Mechanism–controller co-design with stability guarantees · variable-stiffness limbs for manipulation and locomotion · nonlinear and adaptive control · control under actuation lag · fault-tolerant aerial vehicles

Education

International Institute of Information Technology, Hyderabad Nov 2021 – Dec 2026 (expected)
Dual Degree: B.Tech ECE + MS by Research in Robotics *MS coursework GPA 9.0/10* | *Integrated CGPA 7.91/10*
Advisor: Prof. Harikumar Kandath

Coursework: Robot Dynamics & Control, Advances in Robotics and Control, Mobile Robotics, Mechatronics System Design, Intro. to UAV Design, Reinforcement Learning, Statistical Methods in AI

Publications

- **A. Parkala** and H. Kandath, "Spinning Quadrotor: Hover Thrust Augmentation with Passive Lifting Surfaces," in *Proc. Int. Conf. on Unmanned Aircraft Systems (ICUAS)*, 2026. **Best Paper Award Finalist.** DOI | project page | code
- **A. Parkala** and H. Kandath, "Characteristics and Fault-Tolerant Capabilities of Winged Spinning Quadrotors," *IEEE/ASME Trans. Mechatronics*, in preparation.

Research Experience

Research Assistant, Robotics Research Center, IIIT Hyderabad May 2023 – Present
On-board control of a spinning quadrotor at high spin rates *IEEE/ASME T-Mech (in preparation)*

- Identified **actuation lag as the limit on spin rate**: removing the rotor pole raised the simulated ceiling from 22 to $\geq$29.5 rad/s; measured 28–31 ms effective lag from flight logs.
- Developed a **lag-compensated controller** (Lyapunov analysis): an **adaptive phase compensator** plus **rotor-dynamics inversion** hold hover at **29.5 rad/s** in PX4 SITL (30 ms lag); neither alone does.
- Integrated the controller into **PX4** (custom control allocation, spin-invariant pilot modes, rotor-failure injection) and corrected **EKF2 divergence under rapid rotation**.
- Designed and prototyped the airframe and wings solo; sized the wings for lift in the spinning regime with BEMT, CFD and XFLR5.
- Built an **Isaac Sim/Pegasus + PX4 SITL** sim-to-real pipeline; identified dynamics from logs and achieved **controlled spinning flight at 15 rad/s** on hardware (Kakute H7, motion capture).

Hover thrust augmentation with passive lifting surfaces *ICUAS 2026, Best Paper Finalist*
Designed passive wings that carried 22% of vehicle weight on hardware during steady spin at 8–10 rad/s.

Teaching Experience

Teaching Assistant, Perception and Planning for Aerial Vehicles (NPTEL) Jul 2025 – Sep 2026
Developed all slides, assignments and quizzes for three faculty; gave 3 hours of tutorials.

Teaching Assistant, Introduction to UAV Design, IIIT Hyderabad Jan – May 2025
Taught PX4, ROS 2 and XFLR5 to undergraduate, Master's and PhD students.

Teaching Assistant, Digital Systems and Microcontrollers, IIIT Hyderabad Aug – Dec 2023, 2024
Ran Arduino and digital-circuits labs for 340+ students across two offerings.

Selected Projects

Morphology–Policy Co-Design of Compliant Quadruped Legs *Isaac Lab, PyTorch; in progress*
Co-optimising leg length and leg stiffness (CMA-ES) with an RL gait policy; testing how the optimal stiffness shifts with payload.

Adaptive Sliding-Mode Control for Bipedal Locomotion *ROS 2, Gazebo*
Joint-space PID and effort-based adaptive sliding-mode gait control under model uncertainty, evaluated in simulation.

CanSat 2024, International Competition 2023 – 2024
Designed the mechanical subsystems and structure; 4th worldwide at Preliminary Design Review, 30th in world finals.

Awards

Best Paper Award Finalist, ICUAS 2026 · 4th worldwide at the CanSat 2024 PDR · KVPY Fellow, 2019–20

Technical Skills

Control	Nonlinear, adaptive, sliding-mode, LQR, backstepping, Lyapunov/ISS, dynamic inversion, system ID
Flight stacks	PX4 (controllers, EKF2, allocation, uORB), PX4 SITL, ROS 2, MAVLink, ULog analysis
Simulation	Isaac Sim / Pegasus, MuJoCo, Gazebo, ANSYS Fluent, XFLR5
Programming	C/C++, Python (PyTorch), MATLAB, Bash, Git, Linux
Hardware	KakuteH7, ESP32, motors/ESCs, motion capture, PCB design, Verilog, SPICE, Fusion 360